Name: _____

WORKSHEET

Necessary and sufficient

1 Identify whether these statements are 'if and only if' or 'necessary but not sufficient'.

a All frogs are amphibians.

b All triangles have three sides.

c A quadrilateral is a plane shape with an interior angle sum of 360°.

d All metals conduct electricity.

e An even number is divisible by 2.

f A lottery winner must buy a ticket.

2 Explain why each proof below is incorrect.

a All children are younger than their parents.
Sarah is younger than Tess.
∴ Tess is Sarah's parent.

b All blondes have blue eyes.
Chelsea has blonde hair.
∴ Chelsea has blue eyes.

c Regular shapes have all sides equal in length.
A rhombus has all sides equal in length.
∴ A rhombus is a regular shape.

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d All flowers come from plants.

A conifer is a plant.

$\therefore$ Conifers must have flowers.

e All quadrilaterals are symmetric.

A kite is a quadrilateral.

$\therefore$ Kites are symmetric.

f Adding together two prime numbers always gives an even number.

$2 + 1 = 3$

$\therefore$ 3 is an even number.

3 For each of these proofs, identify if it is valid or invalid, giving a reason.

a A parallelogram has two pairs of opposite sides that are parallel and equal in length.

A rectangle has two pairs of opposite sides that are parallel and equal in length.

$\therefore$ A rectangle is a parallelogram.

b All transcendental numbers are irrational.

$\sqrt{2}$ is irrational.

$\therefore \sqrt{2}$ is transcendental.

c The volume of a cylinder is calculated by multiplying the area of its base by its height.

The volume of an object X is calculated by multiplying the area of its base by its height.

$\therefore$ Object X is a cylinder.

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d The square of a number is always an even number.

36 is the square of 6

$\therefore$ 36 is an even number.

e All triangles can have a maximum of one obtuse angle.

Plane shape *Y* has two obtuse angles.

$\therefore$ Plane shape *Y* cannot be a triangle.

f Similar figures have all corresponding sides in the same ratio.

Figures *A* and *B* have all corresponding sides in the same ratio.

$\therefore$ Figures *A* and *B* are similar.

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Answers

1 **a** necessary but not sufficient

b if and only if

c necessary but not sufficient

d necessary but not sufficient

e if and only if

f necessary but not sufficient

2 **a** Conclusion is invalid because the original premise is necessary but not sufficient. It is not necessarily true that somebody who is younger than somebody else is their child.

b Conclusion is invalid because the original premise is false.

c Conclusion is invalid because the original premise is necessary but not sufficient. It is not necessarily true that a shape with all sides equal in length is regular.

d Conclusion is invalid because the original premise is not true in reverse. All plants do not necessarily have flowers.

e Conclusion is invalid because the original premise is false.

f Conclusion is invalid because the original premise is false.

3 **a** Valid; both statements and conclusion are valid and true.

b Invalid; the original premise is false.

c Invalid; the original premise is necessary but not sufficient. Other objects may have their volume calculated with the same formula.

d Invalid; the original premise is false.

e Valid; both statements and conclusion are valid and true.

f Invalid; the original premise is necessary but not sufficient. Similar figures also need to have corresponding angles equal in size and shapes may exist that have all corresponding sides in the same ratio but not have corresponding angles equal in size.

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